INCH-POUND

MIL-PRF-19759F
2 September 1997
SUPERSEDING
MIL-C-19759E
12 December 1989

PERFORMANCE SPECIFICATION

CLOTH, WATERPROOF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for two types of waterproof cloth.

1.2 Classification. The following types of cloth cited in this specification are as follows (see 6.2).

Type I - Green 3410 face, Black back.
Type II - Green 34096 face, Black back.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 4 and 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 4 and 5 of this standard, whether or not they are listed.

2.2 Government documents.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Defense Personnel Support Center, Clothing and Textiles Directorate, Attn: DPSC-FNS, 2800 South 20th Street, Philadelphia, PA 19145-5099.

AMSC N/A **FSC 8305**

DISTRIBUTION STATEMENT A. ***Approved for public release; distribution is unlimited.***

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:38Z
Check the source to verify that this is the current version before use.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods
5206 - Stiffness of Cloth (Drape & Flex)
5556 - Dimensional Change in Laundering (Mobile Laundry Method)
5804 -Weathering of Cloth: Accelerated Weathering Method
5850 -Accelerated Aging of Cloth: Oven Method

FED-STD-595 - Colors Used In Government Procurement

DEPARTMENT OF DEFENSE

MIL-STD-1487 - Glossary of Cloth Coating Imperfections

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk. Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

FEDERAL TRADE COMMISSION

Rules and Regulations Under-the Textile Fiber Products Identification Act

(Copies are available from the Federal Trade Commission, Pennsylvania Avenue at Sixth Street, N.W., Washington, DC 20580-0001.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D-751 - Water Resistance of Cloth; Water Permeability; Hydrostatic Pressure Method
ASTM-D-1424 - Tear Resistance of Woven Fabrics by Falling Pendulum (Elmendorf) Apparatus
ASTM-D-5034 - Weight of Textile Materials: Small Specimen Method
ASTM-D-3776 - Mass Per Unit Area (Weight) of Woven Fabric

(Applications for copies of referenced documents should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19426-2959.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:38Z
Check the source to verify that this is the current version before use.

TECHNICAL MANUAL OF THE AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC - 8 - Crocking

(Applications for copies of referenced documents should be addressed to the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709-2215.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2 (see 6.3).

3.2 Standard sample. The waterproof cloth shall match the standard sample for shade and appearance and shall be equal to or better than the sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

3.3 Materials.

3.3.1 Dusting powder. If dusting powder is used on the cloth, it shall be whiting, talc, or other finely divided mineral material that does not support mildew growth. Talc shall not contain asbestos and whiting shall contain less than 1% quartz. Only the minimum amount of dusting powder necessary to prevent blocking should be applied.

3.4 Performance requirements. The cloth shall conform to the requirements specified in Table I and 3.4 through 3.4.2. If dusting powder is used, the side(s) of the cloth containing the dusting powder shall show not less than "good" resistance to dry crocking.

TABLE I. Requirements

Characteristic	Requirements	
	Type I	Type II
Weight Oz/yd^2 (Max)	9.0	10.5
Breaking Strength, lbs. (Min)		
Warp	225	225
Filling	210	210
Tearing Strength, grams (Min)		
Warp	4500	3000
Filling	3800	3000

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:38Z
Check the source to verify that this is the current version before use.

TABLE I. Requirements (continued)

Characteristic	Requirements Type I	Type II
Hydrostatic Resistance, psi, (Min)		
Initial	100	100
After laundering	80 1/	80 1/
After weathering	80 1/	80 1/
Flex stiffness, inch-lbs X 10^4 (Max)		
At 70°F:		
Warp	9.0	14.0
Filling	7.0	11.0
At 40°F:		
Warp	25.0	35.0
Filling	25.0	35.0
Adhesion of coating, lbs/2 inch width (Min) 2/	10	10
Resistance to heat	3/	3/
Blocking, scale rating (Max)	No. 1	No. 2

1/ The cloth shall show no evidence of cracking, stiffening, or, if finished, blistering or separation of the finish.
2/ Requirement applies if a film or coating is applied to the surface of the cloth.
3/ No evidence of tackiness, exudation, or loss of flexibility.

3.4.1 Color

3.4.1.1 Type I. The color of the cloth shall be Green 3410 on the face side and shall be Black on the back side. (see 3.2 and 6.3).

3.4.1.2 Type II. The color of the cloth shall be Green, No. 34096 of FED-STD-595, on the face side and shall be Black on the back side (see 3.2 and 6.3).

3.4.2 Toxicity. The cloth shall not be toxic to the skin, eyes or epidermis when used as intended (see 4.4.7).

3.5 Fiber identification. Each roll shall be labeled and ticketed for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.6 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Quality conformance inspection (see 4.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:38Z
Check the source to verify that this is the current version before use.

4.2 First article inspection. When a first article is required (see 6.2) it shall be examined for the defects listed in 4.4.1 and shall be tested for the characteristics specified in Table III.

4.3 Quality conformance inspection. Sampling for inspection shall be as specified in the contract or purchase order.

4.3.1 End item examination.

4.3.1.1 Yard by yard examination. During the yard by yard examination, each roll in the sample shall be examined for the defects listed below. However, the side of the cloth examined shall be alternated on every other roll. The same yardage shall be given a through-lighting inspection for pinholes only in accordance with MIL-STD-1487. All defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. The sample unit shall be one linear yard. An approximate equal number of yards shall be examined from each roll in the sample.

Defect	Classification	
	Major	Minor
Type I cloth, face side:		
Any hole, cut or tear	101	
Smash, i.e., damaged area where the texture has been ruptured	102	
Baggy, ridgy or wavy	103	
Any abrasion mark or bruise, (i.e., area damaged by friction)	104	
Any evidence of strike through of finish (if finished)	105	
Any spot or stain (longest dimension in any direction) - up to and including 1/4 inch. in length - more than 1/4 inch in length	 106	 201
Any evidence of a ridge or pucker	107	
Any evidence of imbedded foreign matter or waste (e.g., specs or fly neps in colors other than that of the cloth	108	
Any nonuniformity in appearance (i.e., broken or missing yarn): - up to and including 1/4 inch in length - more than 1/4 inch in length	 109	 202
Any area of general uncleanliness		203
Type I cloth, face side:		
Any lump (i.e., knot, larger than C or slub larger than D or 3 1/2 on the Sears Fabric Defect Scale (see 6.6)	110	
Off shade, shaded side to side, side to center, end to end, mottled, cloudy or streaked 1/	111	
Type II face side and Types I and II back side:		
Any unfinished area (if finished)	112	
Any cut, deep scratch, heavily abraded area, hole or tear	113	
Any thin area	114	
Any blister, tunnel, or separation of finish (if finished)	115	
Any lump or thick area		204
Crease or wrinkle resulting in doubling or adhesion of surfaces that cannot be corrected by manual pressure	116	
Finish cracked or checked (if finished)	117	
Finish not applied over full width of cloth (if finished)	118	

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:38Z
Check the source to verify that this is the current version before use.

Defect	Classification	
	Major	Minor
Any pinhole	119	
Any transparent area or window	120	
Any evidence of puckering, waviness or distortion	121	
Off shade, shaded side to side, side to center, end to end, mottled, cloudy or streaked 1/	122	
Any area of general uncleanliness		205
Any tackiness, cloth adheres to itself and will not readily unroll	123	
Any objectionable odor 2/		206
Any spot, stain, streak 3/:		
less than 1 inch in its longest direction		207
1 inch or more in its longest direction)	124	
Any imbedded foreign matter	125	
Any brittleness (i.e., finish cracks when flexed (if finished)	126	
Edges folded, rolled, or not cut straight		208

1/ Cloth which has been dusted with dusting powder shall be wiped with a wet cloth and allowed to dry before color matching.
2/ Odors of chemicals or commonly used finishes (i.e., films and coating compounds) shall not be regarded as objectionable.
3/ Clearly visible at normal inspection distance (approximately 3 feet).

4.3.1.2 Roll identification examination. During the yard-by-yard examination, each roll in the sample shall be examined for proper identification. Any roll in the sample that is not labeled or ticketed in accordance with the Textile Fiber Products Identification Act shall be considered a roll identification defect.

4.3.1.4 Shade and appearance examination. During the yard-by-yard examination, each roll in the sample shall be examined on the face side for shade and appearance. If any roll in the sample is off shade or shaded side to side, side to center, or end to end, or if any roll does not have the same appearance as the standard sample, the lot represented by the sample shall be rejected.

4.3.2 End item. The cloth shall be tested for the characteristics listed in Table II. The methods of testing specified wherever applicable and as listed in Table II shall be followed. The lot shall be unacceptable if any sample unit fails to meet any requirement specified. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be 3 continuous yards full width of cloth. The lot shall be unacceptable if any sample unit fails to meet any requirement specified. The lot size shall be expressed in units of one yard. The sample size shall be as follows:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:38Z
Check the source to verify that this is the current version before use.

TABLE II. End item tests

Characteristic	Test method
Resistance to dry crocking	AATCC Method 8 1/
Weight	ASTM D 3776 Method C (small swatch of fabric method)
Breaking strength	ASTM D 5034 (G-E or G-T)
Tearing strength	ASTM D 1424
Hydrostatic Resistance Initial After laundering After weathering	 4.4.1.1 4.4.1.2 4.4.1.3
Flex Stiffness, At 70°F: At - 40°F:	 4.4.2.1 4.4.2.2
Adhesion of coating	4.4.3
Resistance to heat	4.4.4
Blocking	4.4.5
Color	4.4.6
Toxicity	4.4.7

1/ The test shall be conducted using the blue/black crock cloth.

4.4 Methods of inspection.

4.4.1 Hydrostatic resistance.

4.4.1.1 ASTM D-751, Hydrostatic Resistance Procedure A (Pressure Application by Mullen Type Hydrostatic Tester) Procedure 1 with water pressure applied to the face side of the waterproof cloth.

4.4.1.2 The cloth shall be laundered five (5) times in accordance with method 5556 of FED-STD-191, Cotton Laundering procedure. After laundering, the cloth shall be examined for evidence of cracking, stiffening and, if finished, blistering or separation of the finish. The cloth shall then be tested for hydrostatic resistance in accordance with 4.4.1.1.

4.4.1.3 The back side of cloth shall be weathered for eighty (80) hours in accordance with method 5804 of FED-STD-191. After weathering, the cloth shall be examined for evidence of cracking, stiffening and, if finished, blistering or separation of the finish. The cloth shall then be tested for hydrostatic resistance in accordance with 4.4.1.1.

4.4.2 Flex Stiffness.

4.4.2.1. The cloth shall be tested for flex stiffness in accordance with method 5206 of FED-STD-191, except that ten (10) specimens shall be tested and averaged. Five (5) of the 10 specimens shall be tested face side up and the remaining five shall be tested with back side up.

4.4.2.2 The stiffness test shall be conducted as specified in 4.4.2.1 except that the apparatus and test specimens shall be subjected to a temperature of minus 40°F $\pm$ 5°F for a period of one (1) hour and the test shall be performed in a still atmosphere at that temperature.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:38Z
Check the source to verify that this is the current version before use.

4.4.3 Adhesion of coating. ASTM-D-751, Adhesion of Coating with 2 inch wide reinforced coating adhesion specimens, cyanoacrylate (solventless) adhesive, and pulling clamp speed of 5 mm/s. Three specimens of the cloth shall be tested by adhering back-to-back.

4.4.4 Resistance to heat. The cloth shall be tested for resistance to heat in accordance with method 5850 of FED-STD-191, except the cloth shall be heated in the oven at 160°F ± 2°F for 120 hours and then visually examined.

4.4.5 Blocking. The cloth shall be tested to determine blocking resistance in accordance with ASTM D-751, Determination of Blocking Resistance of Fabrics Coated with Rubber or Plastics at Elevated Temperatures, except that the test specimens shall be exposed to an oven temperature of 180°F ± 2°F for 30 minutes.

4.4.6 Color matching. The color of the cloth shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 ± 200°K, with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200°K.

4.4.7 Toxicity assessment. The waterproof cloth shall be tested for toxicity as follows:

a) Title 40, Code of Federal Regulations, 1994 Edition;
   Section 798.4100- Dermal Sensitization
   Section 798.4470- Primary Dermal Irritation
   Section 798.4500- Primary Eye Irritation
   Marzulli, F. & H. Maibach, "Contact Allergy: Predictive Testing in Humans", Advances in Modern Toxicology, Volume 4, pp 353-372, 1977.
b) As an alternative to animal and human testing, the contractor may provide information which certifies that the finished product was composed of chemicals or materials which have been safely used commercially where prolonged skin contact has occurred.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The type I coated cloth is intended for use in the manufacture of special cold weather clothing items. The type II coated cloth is intended for use in the manufacture of wet weather clothing items.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:38Z
Check the source to verify that this is the current version before use.

6.1.1 Vulcanization. It should be noted that various end items fabricated from these cloths require that the material be furnished in the unvulcanized state. Samples furnished for testing under this document are, however, fully vulcanized prior to testing. The end item procurement documents should therefore be reviewed to determine this requirement. Examples of items that would require that the cloth be furnished unvulcanized are: Parka, Wet Weather, MIL-P-87098 and Trousers, Wet Weather, MIL-P-87099.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When first article is required (see 3.1, 4.2, and 6.3).
e. Width required.
f. Length required.
g. Levels of preservation and packing (see 5.1).

6.3 Standard sample. For access to the standard sample, address the contracting activity issuing the invitation for bids (see 3.2).

6.4 Subject term (key word) listing.

Fabric, weather proof
Clothing, wet weather

Custodians:
Navy - NU
Air Force - 99

Preparing activity
DLA-CT

Review activities:
Air Force - 45, 82

(Project 8305-0613)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:38Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19759F	**2. DOCUMENT DATE** ***(YYMMDD)*** 97 September 02

3. DOCUMENT TITLE

CLOTH, WATERPROOF

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME DEFENSE PERSONNEL SUPPORT CENTER DPSC-FNS	b. TELEPHONE *(Include Area Code)* (1) Commercial (215) 737-8079 (2) AUTOVON 444-8079
c. ADDRESS *(Include Zip Code)* 2800 South 20th STREET PHILADELPHIA, PA 19145-5099	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:38Z
Check the source to verify that this is the current version before use.